GUC
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الجامعة الألمانية بالقاهرة

AI-Driven Pay Equity Analytics: Detecting and Addressing Compensation Disparities

Seminar Paper Proposal

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Use Cases

Use Case 1: View Salary Explanation

- **Actor:** Employee
- **Description:** The system generates a personalized explanation based on a **regression model prediction vs actual salary**.
- **Postcondition:** Human-readable explanation is displayed including fairness status.

Use Case 2: Detect Pay Inequity

- **Actor:** HR Manager
- **Description:** The system detects inequity using a **machine learning model (Linear Regression)** trained on:
  - Experience
  - Performance
  - Role (one-hot encoded)
- **Postcondition:** Employees flagged as **Potential Inequity = True**

Use Case 3: Run Simulation (Salary Adjustment)

- **Actor:** HR Manager
- **Description:** Simulates salary corrections using a **budget allocation algorithm**.
- **Key Logic:**
  - Prioritizes **most underpaid employees first**
  - Adjusts salaries toward predicted fair salary
  - Stops when **budget is exhausted**
- **Postcondition:** Updated salaries + fairness improvement metrics

Use Case 4: Monitor Fairness Dashboard

- **Actor:** Executive
- **Description:** Displays aggregated fairness indicators.
- **Metrics :**
  - Average Salary

- Gender Pay Gap (%)
- Number of flagged employees

- **Postcondition:** Executive insights for decision-making

2. Epics & User Stories

Epic 1: AI-Based Salary Explanation

User Stories

- As an **employee**, I want a breakdown comparing my salary to a **model-predicted fair salary**
- As an **employee**, I want to know if I am underpaid based on a **5% fairness threshold**

Epic 2: Automated Bias & Inequity Detection

User Stories

- As an **HR manager**, I want the system to detect inequity using **data-driven predictions instead of averages**
- As an **HR manager**, I want to identify employees whose salary deviates significantly from expected values

Epic 3: Fair Salary Adjustment Simulation

User Stories

- As an **HR manager**, I want to allocate a fixed budget to fix underpaid employees
- As an **HR manager**, I want the system to prioritize employees with the **highest negative salary gap**

Epic 4: Governance & Fairness Monitoring

User Stories

- As an **executive**, I want to track **gender pay gap %**
- As an **HR manager**, I want recommendations based on detected bias patterns

3. BPMN

Epic 1: Salary Explanation Process

1. Employee requests explanation
2. Retrieve employee record
3. Run regression model prediction
4. Compute Delta %
5. Check if |Delta| > 5%
6. Generate explanation text
7. Display result

Epic 2: Inequity Detection Process

1. Load dataset
2. Train Linear Regression model
3. Predict fair salaries
4. Compute Delta %
5. Decision: |Delta| > 5%?
    a. Yes → Flag as inequity
    b. No → Continue
6. Aggregate bias metrics (gender pay gap, delta by gender)
7. Generate alerts & recommendations

Epic 3: Simulation Process

1. Input budget value
2. Identify underpaid employees (Salary < Predicted)
3. Sort by most underpaid (lowest Delta %)
4. Loop:

a. Allocate budget to employee
b. Adjust salary
c. Reduce remaining budget
5. Recalculate fairness metrics
6. Generate comparison report

Epic 4: Governance Process

1. Compute fairness metrics
2. Compute gender pay gap
3. Detect bias signal (difference in average Delta %)
4. Generate recommendations:
    a. Salary adjustments
    b. Bias review
    c. Role-based investigation
5. Display dashboard

4. Test Cases

TC-01: Salary Prediction Accuracy

- **Objective:** Verify model generates predicted salary
- **Expected Result:** Predicted_Fair_Salary is calculated for all employees

TC-02: Inequity Threshold Detection

- **Objective:** Validate 5% rule
- **Steps:**
    - Input employee with >5% deviation
- **Expected Result:** Potential_Inequity = True

TC-03: Gender Pay Gap Calculation

- **Objective:** Verify formula
- **Expected Result:**

*(Maleavg−Femaleavg)/Maleavg∗100(Male_{avg} - Female_{avg}) / Male_{avg} * 100*(Maleavg −Femaleavg)/Maleavg ∗100

TC-04: Budget Allocation Simulation

- **Objective:** Verify fairness improvement
- **Steps:**
  - Set budget
  - Run simulation
- **Expected Result:**
  - Budget decreases correctly
  - Most underpaid employees adjusted first

TC-05: Bias Detection Signal

- **Objective:** Detect systematic bias
- **Expected Result:**
  - If female delta << male delta → bias flagged

TC-06: Recommendation Engine

- **Objective:** Validate recommendations
- **Expected Result:**
  - Suggestions generated based on:
    - Pay gap
    - Inequity count
    - Bias signal